Process Characterization Report

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCR-003

Process Development / Manufacturing Science & Technology (MSAT)

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Abbreviations

2-AB, 2-aminobenzamide; ADCC, antibody-dependent cell-mediated cytotoxicity; AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; Cpk, one-sided process capability index; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HILIC, hydrophilic interaction liquid chromatography; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; pCO_2, dissolved carbon dioxide partial pressure; PPQ, process performance qualification; QbD, quality by design; qPCR, quantitative polymerase chain reaction; RSM, response-surface methodology; SD, standard deviation; SDM, scale-down model; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; UPLC, ultra-performance liquid chromatography; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

The production bioreactor is the step at which A-Mab acquires its glycan, charge-variant and aggregate quality attributes. No later step in the drug-substance process modifies the glycan profile, so the operating region defined here fixes those attributes for the product. This report is the record of the characterization study executed under PCP-003 in a qualified scale-down model, and it supports the transfer of the drug-substance process from Cambridge to Grafton (PTP-001). The approach follows the lifecycle model for process validation (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8 and ICH Q11, with criticality treated as a continuum in the manner of the A-Mab case study (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b).

9 process parameters were characterized, of which 5 were studied together in a screening design of 19 runs and a response-surface design of 28 runs, while the remaining 4 were assessed one at a time. 5 parameters are classified as well-controlled critical process parameters, 3 as key process parameters and 1 as a general process parameter. No parameter at this step required the most stringent classification (§9).

The step sets 7 quality attributes, 4 of which are critical (Table 2.1), and five of them were measured as responses in the designed experiments. The response-surface models are adequate for all five, with coefficients of determination from 0.91 to 0.97, an adjusted value no lower than 0.81, and no significant lack of fit against the centre-point pure error (Table 5.3). Those models, and not the screening fit, are the basis of the design space in §6.

One corner of the characterized region is excluded from the design space. Where culture pH, dissolved carbon dioxide and culture duration are all at their low edges and temperature is at its high edge, galactosylation is predicted at 40.3 %, which is above its upper limit of 40.0 %. The exceedance was observed as well as predicted, in 1 run of the screening design and 1 run of the response-surface design. The corner lies far outside the normal operating ranges, and within those ranges the worst predicted galactosylation is 31.8 %, leaving a margin of 8.2 percentage points to the limit.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches operated within the normal operating ranges. Every attribute formed at this step meets its acceptance criterion. The tightest capability is 1.94 for High Mannose, whose mean sits 2.99 percentage points above its lower limit; the widest is 16.1 for Aggregates (HMW).

2 deviations were recorded during execution. Both were investigated, impact-assessed and retained, and neither altered a fitted effect, a parameter classification or the boundary of the operating region (§13).

The step does not clear any impurity, since host cell protein and residual DNA are generated by the culture and removed downstream, with their drug-substance capability credited to Protein A chromatography (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008). No viral clearance is claimed for the production bioreactor. The results of this study roll up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose principal mechanism of action is antibody-dependent cell-mediated cytotoxicity (ADCC). Two of its glycan attributes, afucosylation and galactosylation, modulate that activity directly, and a third, high mannose, is linked to clearance. The drug substance is produced by fed-batch culture and purified on a platform train of Protein A chromatography, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and ultrafiltration with diafiltration (Steps 5 to 10).

The production bioreactor is the third step of that train. Cells from the N-1 seed bioreactor are inoculated into basal medium, and the culture is fed and controlled to fixed set-points for pH, temperature, dissolved oxygen and dissolved carbon dioxide until harvest (SOP-2003). The glycan attributes are formed by the glycosyltransferase and glycosidase activities of the cell during that culture, so they are set by the culture environment and by its duration. Aggregate and charge variants are formed by the same culture, principally through the time the product spends in the vessel at the prevailing pH and temperature.

The step's role in the control strategy follows from this, because downstream steps clear impurities and viruses but the platform purification does not modify the distribution of glycosylation variants, and the charge-based steps do not discriminate between them (CMC Biotech Working Group 2009). The quality attributes formed here are therefore controlled here, and the operating region defined in §6 is the principal control for them.

7 quality attributes are attributed to this step. Five of them are formed in the culture and were measured as responses in the designed experiments. The other two, host cell protein and residual DNA, are process impurities that the culture generates and that later steps remove.

1.2 Objectives

The objectives of the study were:

(i) to quantify the relationship between each characterized process parameter and the quality attributes formed at this step, over ranges wider than the routine control ranges;

(ii) to define a multivariate operating region within which those attributes meet their acceptance criteria;

(iii) to establish proven acceptable ranges for each parameter against each governed attribute;

(iv) to classify each parameter according to its demonstrated effect and the risk of operating outside the region; and

(v) to justify the Stage 2 operating ranges and the contribution of this step to the control strategy for the drug substance.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the lifecycle model for process validation (U.S. Food and Drug Administration 2011). It follows the enhanced development approach described in ICH Q8 and ICH Q11, in which process understanding is built from prior knowledge and designed experiments, and a design space is proposed on the strength of that understanding (International Council for Harmonisation 2009, 2012). Criticality is treated as a continuum in the sense of ICH Q9 and of the A-Mab case study, so attributes are assigned a level of criticality and those at the high and very high levels are called critical (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009).

No viral clearance claim is made for this step, because the viral safety of the drug substance rests on the modular clearance evaluated at low-pH inactivation (PCR-006), anion exchange (PCR-008) and small-virus retentive filtration (PCR-009), and is consolidated in PCMR-001 under ICH Q5A (International Council for Harmonisation 2023a).

The documents that frame and consume this report are listed in Table 1.1.

Table 1.1: Related documents in the characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-003	Process Characterization Plan (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The risk assessment RA-001 set the scope of the study, PCP-003 is the approved protocol it was executed against, and PCMP-001 is the master plan for the campaign. Any departure from PCP-003 is reported in §13.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The A-Mab cell culture process is a platform fed-batch process that has produced three previous humanized IgG1 antibodies, X-Mab, Y-Mab and Z-Mab, as well as the A-Mab clinical and engineering material. The medium, feed strategy, control set-points and vessel design are common to those programmes. The behaviour of the platform is therefore known, and the purpose of this study was to confirm and bound that behaviour for A-Mab, not to discover it.

Prior knowledge sets a mechanistic expectation for each parameter, and the study was designed to test that expectation. Culture pH governs the intracellular pH of the Golgi and the activity of the glycosyltransferases, so it is expected to move all three glycan attributes (afucosylation, galactosylation and high mannose). Culture temperature acts on specific productivity and on the balance between growth and production, and it is expected to move galactosylation, although prior knowledge does not fix the direction of that effect for this cell line. Dissolved carbon dioxide accumulates during fed-batch culture and lowers the intracellular pH, so it is expected to act on galactosylation and on deamidation in the same direction as culture pH. Osmolality rises with feeding and acts on specific productivity. Culture duration sets the total residence time of the product in the vessel, so it is expected to raise aggregate and to lower galactosylation through the accumulation of extracellular glycosidase activity.

Two of these expectations were carried forward from X-Mab, where the acceptance limits for afucosylation and galactosylation were themselves derived from clinical experience. That lineage is the reason the limits are treated as fixed inputs to this study and not as outcomes of it.

2.2 Quality attributes in scope

The quality attributes primarily set by this step are given with their acceptance criteria and assigned criticality (Table 2.1).

Table 2.1: Quality attributes set at the production bioreactor, with acceptance criteria, criticality and the A-Mab Tool #1 score.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60

CQA	Category	Acceptance	Criticality	Tool #1
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6

4 of the 7 attributes are of high or very high criticality and are therefore critical. High Mannose carries the highest Tool #1 score in the table, at 80, because its impact on clearance is well established and the uncertainty around that impact is low. Afucosylation and galactosylation are of high criticality because both modulate the effector function on which the mechanism of action depends. Aggregate is of high criticality on immunogenicity grounds. Acidic charge variants are of very low criticality for A-Mab, and residual DNA likewise, but both were retained as responses or as monitored attributes because a large change in either would indicate a change in the culture that other measurements might miss.

The two process impurities in Table 2.1 are treated differently from the five formed attributes. Host cell protein and residual DNA are generated by the culture and are cleared by the purification train, so the bioreactor sets their input level and not their drug-substance level. The clearance credit is given in PCR-005, PCR-007 and PCR-008, and the cumulative position is consolidated in PCMR-001.

2.3 Risk-based prioritization and parameter selection

The pre-characterization risk assessment RA-001 ranked every parameter of this step on its potential impact on the quality attributes and on its potential to interact with other parameters. The ranking was used to sort the parameters into those warranting multivariate evaluation, those whose ranges could be supported by univariate work, and those requiring no new study. Where no data or rationale were available to support an assessment, the parameter was ranked at the highest level, which is the conservative default used throughout the campaign.

5 parameters were assigned to the multivariate study on that basis: culture pH, culture temperature, dissolved carbon dioxide, osmolality and culture duration. Each of them has a direct mechanistic route to at least one glycan or charge-variant attribute, and each was scored as a likely participant in interactions. The remaining 4 parameters

were assigned to univariate assessment because their expected route to product quality is indirect and runs through cell growth.

The characterized ranges were set two to three times wider than the routine control ranges (Table 4.1), as in the A-Mab case study (CMC Biotech Working Group 2009). The intent was to establish process understanding across a region large enough to support the operating ranges with margin, to allow movement within the design space later in the lifecycle, and to make an edge of failure detectable if one existed inside a plausible operating envelope. Each range was also bounded by what the vessel can hold and by what the culture tolerates. Culture pH was studied over 6.60 to 7.10 because control below the lower bound suppresses growth on this platform. Culture duration was studied to 19 days because viability on the platform falls steeply beyond that point. Dissolved carbon dioxide was given the widest relative range, from 40 to 160 mmHg, because it is the parameter least well controlled at commercial scale and the one most likely to differ between the scale-down model and the 15,000 L vessel.

3 Materials and methods

3.1 Scale-down model and its qualification

The characterization runs were executed in the qualified scale-down bioreactor model described in SOP-1001 and operated under SOP-2003. The model is a stirred-tank bioreactor of the same design family as the commercial vessel, with equivalent design characteristics (mixing, aeration and mass transfer) and equivalent process control capability (pH, temperature, dissolved oxygen and nutrient addition).

Qualification followed the two-part convention used across the campaign, in which scale-independent variables (culture pH, temperature, dissolved oxygen and culture duration) were operated at the proposed target values of commercial operation. Scale-dependent variables (agitation, gas flow rates, headspace pressure, working volume and the resulting dissolved carbon dioxide) were set at small scale so that the culture experiences the same profile as at 15,000 L. Qualification then compared growth, viability, titer and the five product-quality responses between the model and the at-scale reference batches, and found no difference that the analytical methods could resolve.

The warrant this gives the study is specific and bounded. The model reproduces the culture environment that forms the quality attributes, so parameter-response relationships measured in it transfer to the commercial vessel. It does not reproduce commercial-scale mixing heterogeneity at full working volume, and it cannot demonstrate the reproducibility of a commercial batch. Those two questions are answered by the process performance qualification campaign and not by this report.

The centre-point replicates of each design provide the pure-error term against which the fitted models are tested for lack of fit. Their reproducibility is reported below (§5.1), and it is also the practical measure of how well the scale-down model repeats.

3.2 Cell line, media and culture operation

Each run used one vial of the A-Mab working cell bank, expanded through the platform seed train to the N-1 seed bioreactor and inoculated into the characterization vessel at the target initial viable cell concentration. Basal medium and nutrient feeds were prepared under SOP-2004 from the same lots across the campaign, except where noted below (§13). Feeds were delivered on the platform schedule as a percentage of working volume.

Inputs controlled to platform specification were not characterized in this study, because the cell bank, the basal medium formulation, the feed formulation and the gas supply are identity-controlled and quality-controlled materials whose variability is managed by the material specifications and by SOP-2004. Only parameters that the operator sets, and that can move within a control range during manufacture, were treated as characterization factors.

The culture was terminated at the scheduled duration for the run, and the harvest was clarified to provide the material analysed for product quality. Clarification itself is characterized in PCR-004 and forms no quality attribute.

3.3 Analytical methods

The controlled documents governing the study and its analytical methods are listed in Table 3.1.

Table 3.1: Controlled documents and validated analytical methods cited by this report.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

Each response is measured by one validated method, and the three glycan responses are measured together on the released N-glycan map by HILIC-UPLC under SOP-3010 (validated as AMV-3010). Aggregate is measured as high molecular weight species by SEC-HPLC under SOP-3011 (AMV-3011). Acidic charge variants are measured by imaged capillary isoelectric focusing under SOP-3013 (AMV-3013). Host cell protein is measured by ELISA under SOP-3012 (AMV-3012) and residual DNA by qPCR under SOP-3014 (AMV-3014).

Two features of the methods matter for the interpretation of the results. The N-glycan map has a relative precision of 2.5 % and a limit of quantitation of 0.5 % of glycans, and it accounts for about 30.0% of the total observed variance in the glycan responses. The host cell protein ELISA is the least precise method in the set, at 9.5 %, and it accounts for about 40.0% of the observed variance in that assay. Method performance is summarized in Appendix D and is reported in full in the validation reports.

3.4 Sampling plan

Product-quality samples were drawn from the clarified harvest of every run, so each design point yields one observation of each response. Growth, viability and metabolite profiles were sampled daily as process monitoring and are not analysed as responses in this report.

The screening design carried 3 centre-point replicates and the response-surface design carried 4. Centre points were distributed through the execution order of each design so that they sample any drift over the campaign as well as run-to-run variation. Run order within each design was randomized under the study analysis procedure (SOP-1002).

3.5 Statistical methods

Factors were coded so that the set-point of each parameter is zero and the edges of its characterization range are minus one and plus one. Coding puts the factors on a common scale, so an effect estimate is the change in the response across the full studied range of that factor, and effects can be compared between parameters with different units.

The screening designs were analysed by ordinary least squares against a model containing all main effects and all two-factor interactions. An effect is reported as twice the coded coefficient, which is the change in the response between the low and high edges of the factor. The response-surface designs were analysed against the full quadratic model in the response-surface factors. Terms were judged significant at the five per cent level, and the analysis followed the campaign procedure for designed experiments (SOP-1002).

Model adequacy was assessed on four grounds: the coefficient of determination and its adjusted form, the predicted coefficient of determination computed from the prediction error sum of squares, the overall model F-test, and an analysis of variance partitioning the residual into lack of fit and pure error. The pure-error term comes from the centre-point replicates, so lack of fit is tested against the reproducibility the model itself must respect. A parameter with no significant effect on any response is treated as a robustness result and is retained in the knowledge space, not discarded.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches, with each parameter drawn across its normal operating range and each response computed from the fitted response-surface model. Capability is reported as a one-sided index against the binding limit of each attribute, and the index is read together with the absolute margin to that limit. No fixed acceptance threshold for the index is claimed in this report.

4 Study design

4.1 Factors, ranges and the knowledge space

The design resolves the effects of the five parameters that RA-001 ranked highest, and then quantifies the behaviour of the four that matter most over a region large enough to define an operating space. The parameters, their set-points, their normal operating ranges, their characterization ranges and their final classification are given in Table 4.1.

Table 4.1: Process parameters of the production bioreactor, with set-points, normal operating ranges, characterization ranges, final classification and study type.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Culture pH	pH	6.85	6.75–6.95	6.6–7.1	WC-CPP	multivariate
Culture temperature	°C	35	34.5–35.5	34–36	WC-CPP	multivariate
Dissolved CO2 (pCO2)	mmHg	100	70–130	40–160	WC-CPP	multivariate
Osmolality	mOsm	400	375–425	360–440	WC-CPP	multivariate
Culture duration	days	17	16–18	15–19	WC-CPP	multivariate
Dissolved oxygen	% sat	50	40–60	30–70	KPP	univariate
Initial viable cell conc.	e6 cells/mL	1	0.8–1.2	0.5–1.5	KPP	univariate
Basal medium concentration	X	1.2	1–1.4	0.8–1.6	GPP	univariate
Nutrient feed-1 volume	% WV	12	10–14	9–15	KPP	univariate

The characterization range of each parameter is the edge of the knowledge space for this step. The normal operating range sits inside it and is the range the plant is expected to hold. For culture pH the normal operating range spans 6.75 to 6.95, which is 40.0% of the characterized span, and for dissolved carbon dioxide it spans 70 to 130 mmHg, which is 50.0% of it. The wider a characterized range is relative to its operating range, the more margin the study can demonstrate. Culture pH has the most favourable ratio of the five parameters studied multivariately, and osmolality the least at 62.5% of its characterized span.

The coded letters used throughout the effect and coefficient tables are given below (Table 4.2).

Table 4.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	Dissolved CO2 (pCO2)	mmHg	screening + RSM
D	Osmolality	mOsm	screening
E	Culture duration	days	screening + RSM

4.2 Screening design

The screening study was a two-level design in the 5 factors of the legend (Table 4.2), augmented with 3 centre points, for 19 runs in total. Its purpose was to identify which parameters have an effect on which response, and to detect the two-factor interactions that a series of one-at-a-time studies cannot see. The centre-point conditions are the target process conditions. The full design matrix and the responses obtained are given in Appendix A.

The design estimates all 5 main effects and all 10 two-factor interactions, which is 16 model terms from 19 runs. The residual degrees of freedom available to the screening analysis are few, and a two-level design cannot attribute curvature to any individual factor.

4.3 Response-surface design

The response-surface study was a face-centred central composite design in the 4 factors carried forward from screening, augmented with 4 centre points, for 28 runs in total. It adds axial points at the faces of the cube, which allows the quadratic terms to be estimated and the curvature of each response to be measured. The full design matrix and the responses obtained are given in Appendix B.

The division of labour between the two designs is deliberate and is carried through the rest of this report. The screening design identifies which factors are active. The response-surface model quantifies their behaviour, is the model used for prediction, and is the basis of the design space in §6.

Osmolality was not carried into the response-surface design. Its main effects were the smallest of the active factors on both responses where it was significant, and at commercial scale it is held by the feed schedule and not by an independent controller, so the range over which it can move in manufacture is narrow. It does participate in one significant interaction with culture temperature on galactosylation, and the consequence of leaving that interaction out of the response-surface model is stated in §11. Its knowledge-space range is supported by the screening study and by the platform history, and it is classified on that basis in §9.

4.4 Univariate assessment

The 4 parameters not carried into the multivariate work were assessed one at a time across the characterization ranges given in Table 4.1. Their expected route to product quality runs through cell growth, which the culture regulates, and RA-001 ranked all four below the threshold set for multivariate evaluation. The classification that follows from those assessments is given in §9, and the acceptance basis for each is recorded in PCP-003.

5 Results

5.1 Centre-point performance and reproducibility

The centre point of each design reproduced well enough that the fitted models can be tested against it. Centre-point means, standard deviations and relative standard deviations are given for the response-surface design (Table 5.1).

Table 5.1: Centre-point reproducibility of the response-surface design. The standard deviation of each response is the pure-error term used in the lack-of-fit tests.

Response	n	Mean	SD	%CV
Afucosylation	4	7.61	0.355	4.66
Galactosylation	4	25.9	2.34	9.02
High mannose	4	5.66	0.559	9.88
Acidic variants	4	21.5	0.877	4.08
Aggregates (HMW)	4	1.51	0.0985	6.53

The least precise response is High mannose, at 9.9 % relative standard deviation over 4 replicates, and the most precise is Acidic variants at 4.1 %. The same ordering holds in the screening design, where High mannose reproduces at 11.5 %. A large part of that variation is analytical, since the N-glycan map accounts for about 30.0% of the observed variance in the glycan responses, so the culture itself repeats more closely than the numbers in Table 5.1 suggest.

This matters in two places later in the report. The centre-point standard deviations are the pure-error terms in the lack-of-fit tests of §5.3, so a model is only convicted of lack of fit if it misses by more than the assay and the culture together can explain. They also set the resolution of the study, because an effect smaller than about one standard deviation of its response cannot be separated from noise in a single run.

5.2 Screening: factor effects

The screening study resolved the active factors for every response and found interactions between them for four of the five. Model summaries for the screening fits are given in Appendix C (Table 14.5). All five fits have coefficients of determination between 0.94 and 1.00, and the number of terms significant at the five per cent level ranges from 2 for high mannose to 12 for galactosylation.

Galactosylation is the most responsive of the five, and its largest effects are given below (Table 5.2).

Table 5.2: Largest screening effects on galactosylation, in coded units. An effect is the change in the response between the low and high edges of the factor.

Term	Effect	p-value	signif.
duration	-7.82	0.000264	yes
pH × duration	6.9	0.000382	yes
co2	-5.79	0.000644	yes
pH	-4.56	0.00131	yes
temperature × osmolality	3.7	0.00242	yes
pH × co2	3.6	0.00261	yes

Culture duration carries the largest single effect, at -7.8 percentage points across its studied range, and the interaction between culture pH and duration is almost as large at 6.9 points. Dissolved carbon dioxide and culture pH follow, both acting downwards, at -5.8 and -4.6 points. The direction of every one of these agrees with the mechanistic expectation set out in §2.1. Longer culture, higher pCO_2 and higher pH all lower galactosylation, and the presence of a large pH by duration interaction says that the effect of one depends on the level of the other.

Taken together, the screening results establish the multivariate design space for the five well-controlled culture parameters. The effect estimates are large and consistent in direction, and they demonstrate that the step remains robust at parameter combinations beyond those examined here.

Acidic charge variants are governed by dissolved carbon dioxide. Its effect of -8.2 percentage points is the largest single effect anywhere in the screening study, and it acts downwards, which is what raising pCO_2 and lowering the intracellular pH would be expected to do to a deamidation-driven attribute. Culture pH and temperature follow at -4.2 and 3.0 points, and the interaction between them is significant. Culture duration has no effect on this response at all.

Afucosylation and aggregate are governed by fewer factors and by smaller effects. For afucosylation the significant terms are the pH by duration interaction at 2.5 points, culture duration at -2.4 points, and the pH by pCO_2 interaction. For aggregate the significant terms are culture pH at 0.8 points and culture duration at 0.6 points, both acting upwards, which is consistent with aggregate accumulating with residence time and with the product spending longer at the less favourable pH.

High mannose is the simplest response of the five and the most tightly localized, with only culture pH and culture temperature significant, at 1.5 and -1.4 percentage points, and no interaction reaching significance. Dissolved carbon dioxide, osmolality and culture duration have no detectable effect on it over the ranges screened. Only two parameters need to be held closely to govern the attribute of highest criticality at this step, and the three inactive factors are retained in the knowledge space as evidence that high mannose is robust to them across ranges two to three times wider than the operating ranges.

Four culture parameters outside the screening set were held at their set-points for every run of both designs. Dissolved oxygen is the one of these with a credible route to the glycan attributes, because oxygen supply sets the redox state of the culture and the supply of nucleotide sugar donors. Platform manufacturing experience across the humanized IgG1 programmes has established that dissolved oxygen does not influence the glycan profile at any level a production vessel can hold.

Osmolality is significant for galactosylation and for acidic variants, at -2.5 and -2.6 percentage points, and it also participates in an interaction with culture temperature on galactosylation. Both of its main effects are the smallest of the active factors on their respective responses, and across the normal operating range of osmolality they amount to 1.5 and 1.6 percentage points. The interaction with culture temperature on galactosylation is larger than either main effect, at 3.7 points across the studied ranges, and it is the reason the parameter is classified as quality-linked in §9. On that basis osmolality was not carried into the response-surface design, and the magnitudes of the four factors that were carried forward are refined there.

5.3 Response-surface models

The response-surface models are adequate for all five responses and predictive for four of them. Model summaries are given below (Table 5.3).

Table 5.3: Response-surface model summaries. The predicted coefficient of determination is computed from the prediction error sum of squares.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Afucosylation	28	0.963	0.923	0.795	24.3	5.05e-07	0.584
Galactosylation	28	0.971	0.94	0.915	31.3	1.06e-07	1.36
High mannose	28	0.911	0.815	0.65	9.48	0.000118	0.472
Acidic variants	28	0.938	0.872	0.598	14.1	1.27e-05	1.57
Aggregates (HMW)	28	0.969	0.936	0.835	29.3	1.58e-07	0.0985

Coefficients of determination run from 0.91 to 0.97, the adjusted values fall no lower than 0.81, and every model is significant with a p-value no larger than 0.000118. The predicted coefficients of determination run from 0.60 to 0.92, and the lowest of them belongs to Acidic variants. That model remains adequate for interpolation, but its predictions carry visibly more uncertainty than the other four, and the conclusions drawn from it in §6 are correspondingly cautious.

No model shows significant lack of fit. The smallest lack-of-fit p-value across the five responses is 0.15 for Acidic variants, well above the five per cent level, so no model misses the design points by more than the centre-point replicates themselves vary. The analysis of variance for galactosylation, which is the response that governs the operating region, is given in Table 5.4.

Table 5.4: Analysis of variance for the galactosylation response-surface model, with the residual partitioned into lack of fit and pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	811	14	57.9	31.3	1.06e-07
Residual	24.1	13	1.85	nan	nan
Lack of fit	7.69	10	0.769	0.141	0.992
Pure error	16.4	3	5.46	nan	nan

The galactosylation coefficients are given below (Table 5.5).

Table 5.5: Full quadratic coefficients of the galactosylation response-surface model, in coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	25.8	0.47	54.8	9.11e-17	***
A	-2.21	0.321	-6.89	1.09e-05	***
B	2.3	0.321	7.18	7.13e-06	***
E	-3.82	0.321	-11.9	2.27e-08	***
C	-2.43	0.321	-7.56	4.11e-06	***
AB	1.67	0.34	4.92	0.000279	***
AE	2.72	0.34	7.99	2.28e-06	***
AC	2.35	0.34	6.9	1.09e-05	***
BE	-0.217	0.34	-0.639	0.534	
BC	0.315	0.34	0.928	0.37	
EC	0.332	0.34	0.975	0.347	
A^2	-0.837	0.847	-0.988	0.341	
B^2	0.378	0.847	0.446	0.663	
E^2	0.937	0.847	1.11	0.289	
C^2	-0.373	0.847	-0.441	0.667	

Four main effects and three interactions are significant, and no quadratic term is. Culture duration remains the dominant term at -3.82 per coded unit, followed by dissolved carbon dioxide at -2.43 and culture pH at -2.21, all acting downwards, and culture temperature at 2.30 acting upwards. The three significant interactions all involve culture pH, with the pH by duration term the largest at 2.72. A response with three active interactions and no curvature is a tilted plane whose slope in one factor depends on the level of the others, and that shape is what makes a multivariate region necessary.

High mannose behaves differently, and its coefficients are given below (Table 5.6).

Table 5.6: Full quadratic coefficients of the high mannose response-surface model, in coded units.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	5.89	0.163	36	2.06e-14	***
A	1.09	0.111	9.77	2.36e-07	***
B	-0.421	0.111	-3.78	0.00227	**
E	-0.188	0.111	-1.69	0.115	
C	0.212	0.111	1.91	0.0788	
AB	0.148	0.118	1.26	0.231	
AE	-0.29	0.118	-2.46	0.0289	*
AC	0.293	0.118	2.48	0.0276	*
BE	0.0874	0.118	0.74	0.472	
BC	-0.118	0.118	-0.996	0.337	
EC	-0.0402	0.118	-0.341	0.739	
A^2	0.116	0.294	0.393	0.701	
B^2	0.094	0.294	0.32	0.754	
E^2	-0.215	0.294	-0.731	0.478	
C^2	0.126	0.294	0.43	0.674	

Culture pH dominates at 1.09 per coded unit, with culture temperature at -0.42 and two small interactions involving pH. Dissolved carbon dioxide and culture duration are not significant in the response-surface model either, which confirms the screening result over a design that could have detected curvature in them. The coefficient tables for the remaining three responses are given in Appendix C.

The fitted surfaces over culture pH and culture duration are shown in Figure 5.1.

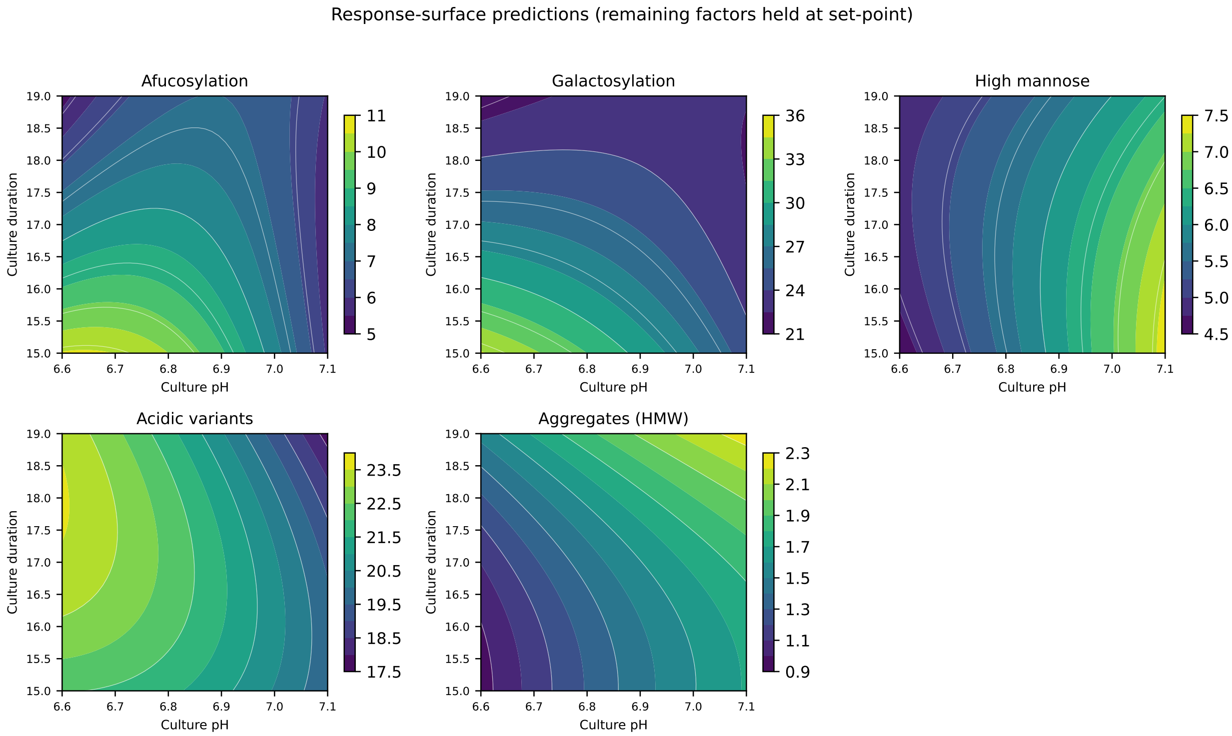

Figure 5.1: Response-surface predictions over culture pH and culture duration, with culture temperature and dissolved carbon dioxide held at their set-points.

The panels show what the coefficients say. Galactosylation falls steeply from the low-pH, short-duration corner towards the opposite corner, and the contours are not parallel to either axis, which is the pH by duration interaction. Aggregate rises with both factors across the whole panel. High mannose depends on pH alone, so its contours are close to vertical. Afucosylation shows the largest curvature in the study, a significant negative quadratic in culture pH of -1.37, which places a maximum inside the studied pH range instead of at an edge.

Model diagnostics for galactosylation are shown in Figure 5.2.

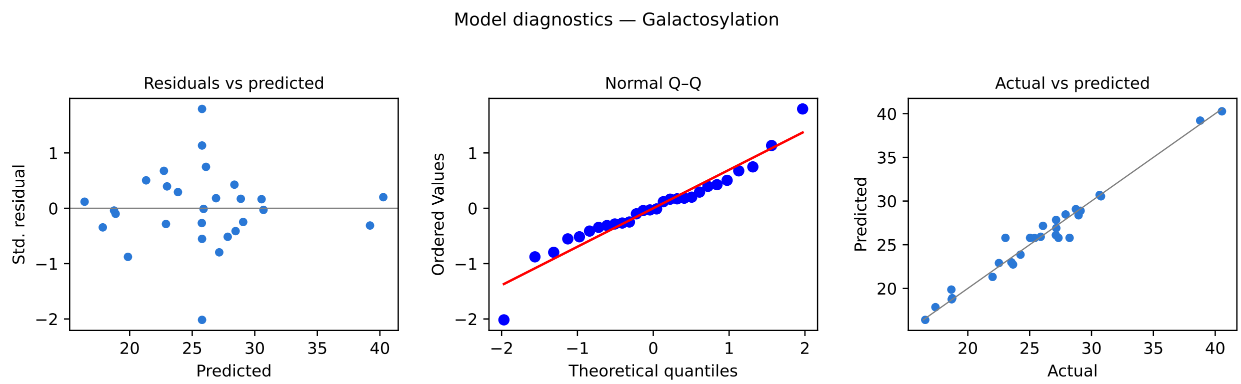

Figure 5.2: Diagnostics for the galactosylation response-surface model: residuals against predicted values, normal quantile plot of the residuals, and observed against predicted values.

The residuals scatter about zero with no trend against the predicted value and no funnel, the quantile plot is close to linear, and the observed values fall on the identity line across the whole range of the response. The model is therefore fit for the prediction of mean levels over the ranges of the parameters included in it, which is the use made of it in §6 and §7.

One prediction of these models is adverse and is reported here. At the corner where culture pH, dissolved carbon dioxide and culture duration are all at their low edges and culture temperature is at its high edge, galactosylation is predicted at 40.3 %, which exceeds its upper limit of 40.0 % by 0.27 percentage points. The exceedance is not only a prediction, because galactosylation was measured above the limit at that same corner in 1 run of the screening design, at 41.4 %, and in 1 run of the response-surface design, at 40.5 %. No other response exceeds any limit anywhere in either design, and at this corner the other four responses are all comfortably inside their criteria. The consequence for the operating region is worked out below (§6).

5.4 Mechanistic interpretation

The surfaces are consistent with the glycan-processing mechanism assumed in §2.1, and the agreement is close enough to treat the models as process understanding and not only as empirical fits.

Galactosylation is a terminal step of N-glycan processing, so the fraction of chains that carry galactose depends on how much processing time the antibody spends in a Golgi with adequate galactosyltransferase activity and adequate donor supply. Everything that shortens or acidifies that window lowers galactosylation. Longer culture lowers it because extracellular glycosidase activity accumulates as viability falls, and the two effects act on the same population of released chains. Higher dissolved carbon dioxide lowers it because carbon dioxide crosses the membrane freely and acidifies the intracellular compartments. Higher culture pH also lowers it, which the study cannot separate from a shift in the pH gradient the cell maintains against the medium. The pH by duration interaction follows directly, because the acidification penalty applies for as long as the culture runs. Higher culture temperature raises galactosylation, at 2.30 per coded unit, which is the direction expected if galactosyltransferase activity limits the reaction and not processing time. Prior knowledge did not fix that direction for this cell line, and the study resolves it.

High mannose is the mirror image of that argument. High mannose glycans are the ones that were never trimmed, so the attribute measures the failure of an early processing step, not the outcome of a late one. Mannosidase activity is the sensitive point, and it responds to intracellular pH and to temperature. Its independence from culture duration is then expected, because trimming happens early in the secretory pathway and the chains that escape it are not recovered later. That independence is the reason the attribute of highest criticality at this step needs the fewest controls.

Aggregate and afucosylation complete the picture. Aggregate rises with culture pH and with duration because both increase the exposure of the product to the extracellular environment, and the interaction between them is not significant, so the two act independently. The only curvature the response shows is a small positive quadratic in duration of 0.143, which says that the last days of culture cost more than the first. Afucosylation has a maximum inside the pH range instead of at an edge, which is the signature of two opposing pH-dependent processes, the supply of the fucose donor and the activity of the fucosyltransferase itself. This is the one response for which the direction of the effect of a pH move depends on where in the range the process is operating.

Two consequences follow for the design of the operating region. All four response-surface factors act on more than one attribute, and they act in opposite directions on some pairs. Raising culture pH lowers galactosylation, raises high mannose and raises aggregate. A univariate range for pH therefore cannot be optimal for all three at once, and the region has to be defined in the parameters jointly. The second consequence is that the interactions are concentrated on culture pH, which makes pH the parameter whose control at scale matters most.

6 Design space

The design space for this step is the region of the four response-surface parameters, bounded by the characterization ranges of Table 4.1, over which the fitted models predict that all five formed attributes meet their acceptance criteria. One corner of the characterized region is excluded from it.

The excluded corner is the galactosylation edge of failure reported in §5.3. It lies at culture pH 6.60, culture temperature 36 °C, culture duration 15 days and dissolved carbon dioxide 40 mmHg. That combination is the worst case the region must satisfy. All four parameters sit at the edge that raises galactosylation, and on balance the interactions between culture pH and the other three add to the effect at that vertex. The exclusion is a small neighbourhood of a single vertex. Holding the other three factors at that vertex, the model returns galactosylation to its limit at a culture pH of 6.617, at a culture temperature of 35.77 °C, at a culture duration of 15.06 days, or at a dissolved carbon dioxide of 44.0 mmHg. Any one of those four moves is sufficient, and each is small against the range of its parameter.

The rest of the characterized region is inside the design space, and the normal operating ranges sit well within it. Evaluating the fitted models at the corners of the normal operating ranges gives a worst predicted galactosylation of 31.8 %, which is 8.2 percentage points below the limit, and every other response meets its criterion at every one of those corners. The distance between the operating ranges and the excluded vertex is large in every factor, and reaching the vertex would require four simultaneous excursions to the edge of the knowledge space.

The principal planes of the region are culture pH against culture duration and culture pH against dissolved carbon dioxide, which carry the two largest interaction terms in the response-surface models. Figure 6.1 shows the second of these.

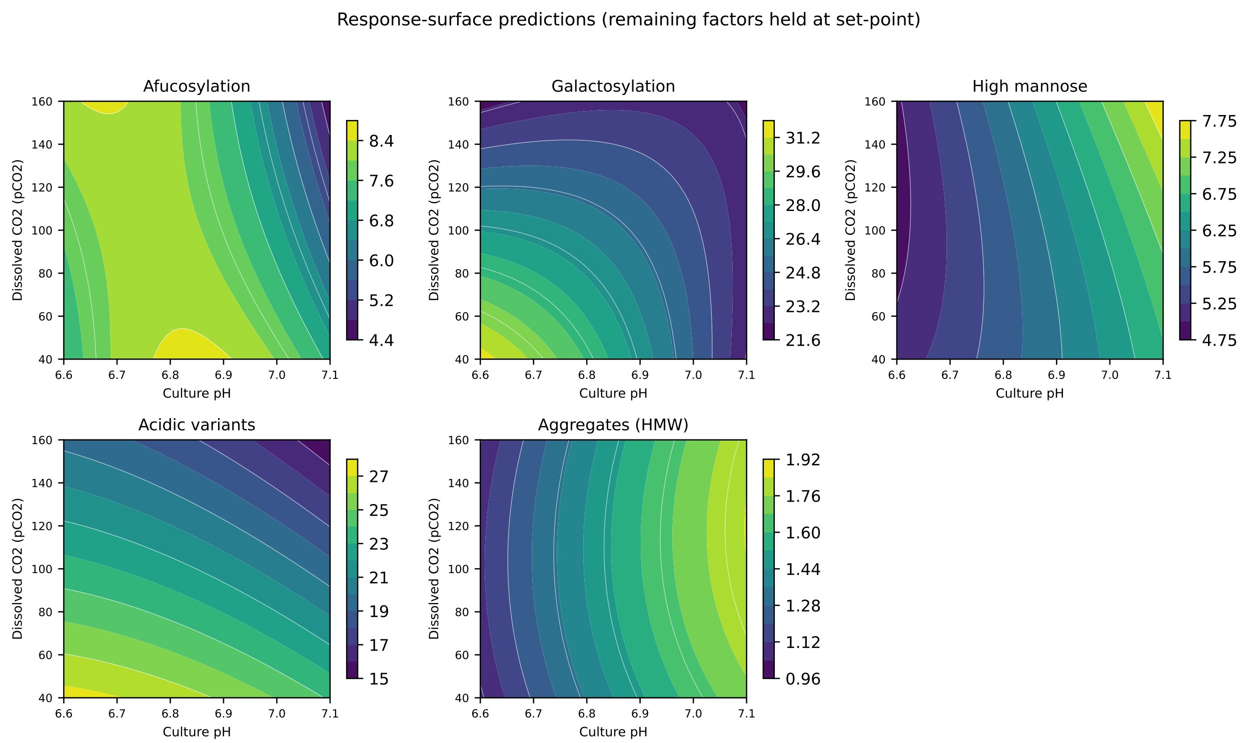

Figure 6.1: Response-surface predictions over culture pH and dissolved carbon dioxide, with culture temperature and culture duration held at their set-points.

Three bounds apply to the claim made in this section. The region is defined over the characterization ranges of Table 4.1 and says nothing about conditions outside them. It is defined by response-surface models of mean levels, so the reliability of a prediction made at the boundary of the region is close to one half if the variability of the response is symmetric about the mean (as in the case study (CMC Biotech Working Group 2009)), and assurance for routine operation comes from the normal operating ranges and the capability of §8 and not from the boundary itself. The region is defined in the qualified scale-down model, and its edges have not been confirmed at commercial scale.

Within the design space, movement is not a change from the perspective of the regulatory filing, although any planned movement is assessed prospectively under the pharmaceutical quality system before it is made (International Council for Harmonisation 2009, 2008). Operation outside the design space, including operation at the excluded vertex, is a change and is managed through change control.

7 Proven acceptable ranges

The design space of §6 is multivariate, and this section complements it with proven acceptable ranges, which are the ranges of a single parameter over which a given attribute meets its criterion while the other parameters are held or varied in a defined way.

The acceptance criterion used for each response is the drug-substance specification for the attribute it maps to, taken from Table 2.1. None of the five responses at this step is a viral-clearance attribute, so no step-level floor has to be back-calculated from a cumulative requirement, and every criterion in this section is a specification limit read directly.

Two analyses were run for each pairing of attribute and parameter, and both are given below (Table 7.1). The first holds the other parameters at their set-points and scans the parameter of interest across its characterization range on the fitted response-surface model. For this step the set-point of every response-surface parameter is also the centre of its characterization range, so the two conventions coincide and the analysis is unambiguous. The second propagates the normal operating ranges of the other parameters by Monte-Carlo sampling of the fitted model, which is the reproducible default across the campaign, and asks over what range of the parameter of interest the attribute still meets its criterion when the others are moving within their routine ranges.

Table 7.1: Proven acceptable ranges for each governed attribute against each response-surface parameter, at set-point and with the other parameters propagated across their normal operating ranges.

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Afucosylation	Culture pH	6.6–7.1	pH	6.6–7.1	6.6–7.1
Afucosylation	Culture temperature	34–36	°C	34–36	34–36
Afucosylation	Culture duration	15–19	days	15–19	15–19
Afucosylation	Dissolved CO2 (pCO2)	40–160	mmHg	40–160	40–160
Galactosylation	Culture pH	6.6–7.1	pH	6.6–7.1	6.6–7.1
Galactosylation	Culture temperature	34–36	°C	34–36	34–36
Galactosylation	Culture duration	15–19	days	15–19	15–19

CQA	Parameter	Char. range	Unit	PAR (set-point)	PAR (NOR)
Galactosylation	Dissolved CO2 (pCO2)	40–160	mmHg	40–160	40–160
High mannose	Culture pH	6.6–7.1	pH	6.6–7.1	6.6–7.1
High mannose	Culture temperature	34–36	°C	34–36	34–36
High mannose	Culture duration	15–19	days	15–19	15–19
High mannose	Dissolved CO2 (pCO2)	40–160	mmHg	40–160	40–160
Acidic variants	Culture pH	6.6–7.1	pH	6.6–7.1	6.6–7.1
Acidic variants	Culture temperature	34–36	°C	34–36	34–36
Acidic variants	Culture duration	15–19	days	15–19	15–19
Acidic variants	Dissolved CO2 (pCO2)	40–160	mmHg	40–160	40–160
Aggregates (HMW)	Culture pH	6.6–7.1	pH	6.6–7.1	6.6–7.1
Aggregates (HMW)	Culture temperature	34–36	°C	34–36	34–36
Aggregates (HMW)	Culture duration	15–19	days	15–19	15–19
Aggregates (HMW)	Dissolved CO2 (pCO2)	40–160	mmHg	40–160	40–160

Every proven acceptable range in Table 7.1 is the whole characterization range of its parameter, under both analyses. No univariate range binds anywhere in the table, and no attribute is brought to its limit by moving one parameter across its full studied range while the others are held at set-point or varied within their operating ranges. Galactosylation against culture duration is the most instructive row, because culture duration carries the largest effect on that attribute and still does not exhaust the margin on its own.

This is the point at which the univariate and multivariate analyses must be read together. Table 7.1 does not see the excluded vertex of §6, and it cannot, because both of its analyses hold the other parameters at or near their set-points while one parameter moves. The vertex is reached only when four parameters move to their edges at the same time. The binding constraint on the operating region is therefore the multivariate corner and not any proven acceptable range, and a control strategy built from Table 7.1 alone would not protect against it. This is the general reason the design space, and not the collection of proven acceptable ranges, is the object that governs operation.

A representative plot is given for each governed attribute against the parameter with its largest response-surface main effect. The acceptable region is shaded, the acceptance limits are drawn, and the set-point and normal operating range are marked on both panels.

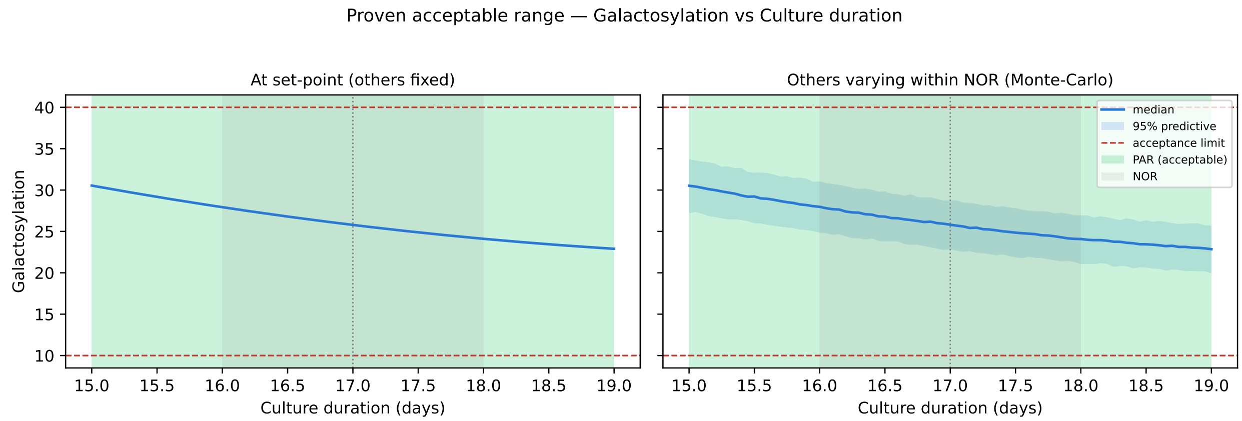

Figure 7.1: Proven acceptable range of culture duration for galactosylation, at set-point (left) and with the other parameters propagated across their normal operating ranges (right).

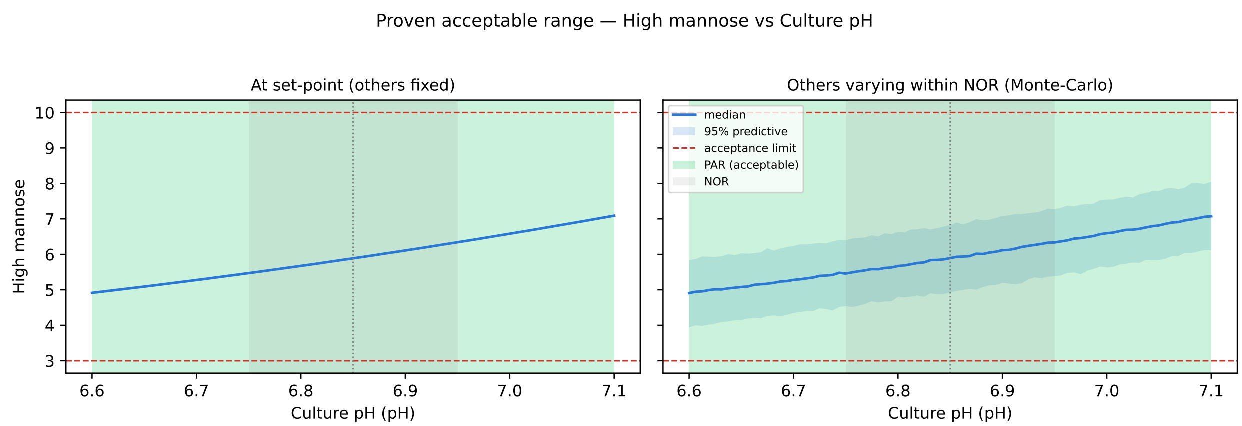

Figure 7.2: Proven acceptable range of culture pH for high mannose, at set-point (left) and with the other parameters propagated across their normal operating ranges (right).

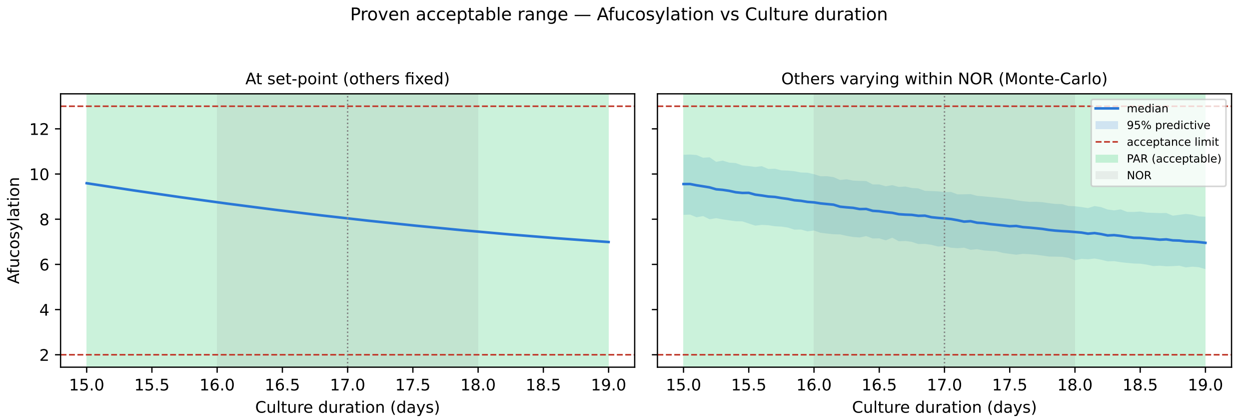

Figure 7.3: Proven acceptable range of culture duration for afucosylation, at set-point (left) and with the other parameters propagated across their normal operating ranges (right).

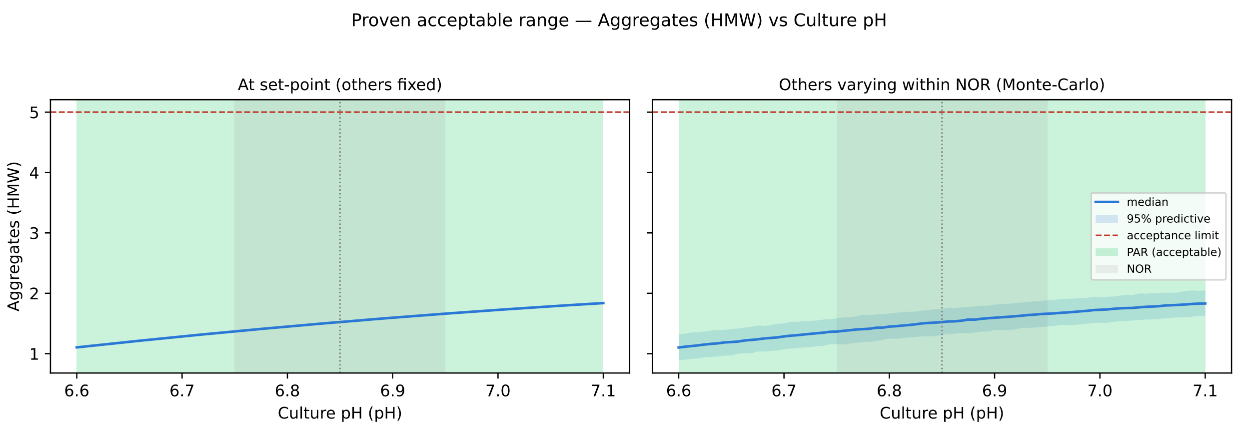

Figure 7.4: Proven acceptable range of culture pH for aggregate, at set-point (left) and with the other parameters propagated across their normal operating ranges (right).

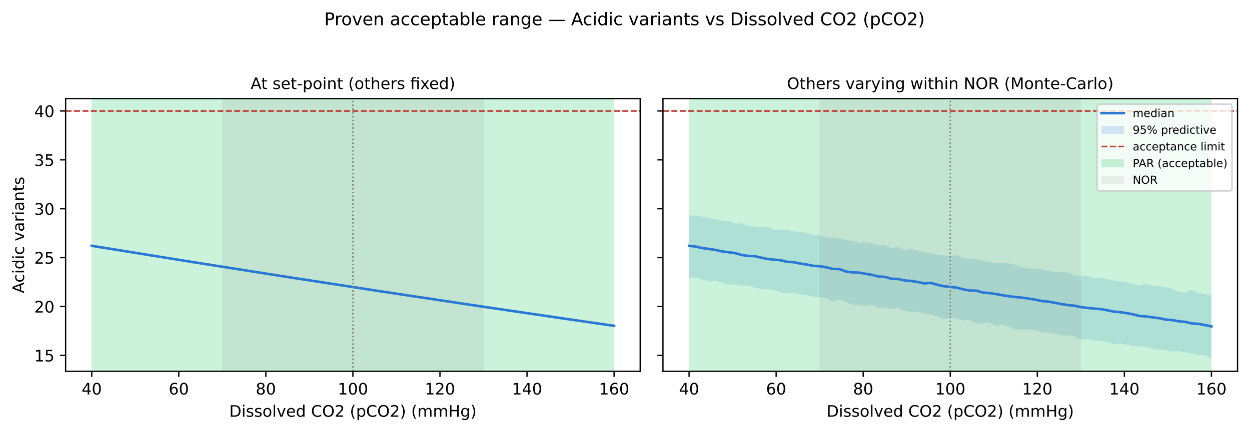

Figure 7.5: Proven acceptable range of dissolved carbon dioxide for acidic charge variants, at set-point (left) and with the other parameters propagated across their normal operating ranges (right).

The plots make the margin visible. In the high mannose panel the response moves towards its lower limit as culture pH falls without reaching it, and the width of the band between the two panels is a measure of how much the other parameters contribute when they move within their operating ranges. That band is narrow for every attribute in this study, which is the same robustness result the capability analysis states numerically in §8.

Two bounds apply to this section. The ranges in Table 7.1 are proven only over the characterization ranges studied, and only under the fitted response-surface models. They are also proven for the attributes formed at this step. Host cell protein and residual DNA are not governed here, and the parameter ranges that assure their drug-substance levels are established at Protein A chromatography (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008).

8 Process capability and robustness

Every quality attribute formed at this step meets its acceptance criterion at commercial scale with margin. Capability was estimated by Monte-Carlo simulation of 2,000 batches with each parameter drawn across its normal operating range and each response computed from the fitted response-surface model, and the results are given below (Table 8.1).

Table 8.1: Commercial-scale capability of the attributes formed at the production bioreactor, from Monte-Carlo simulation of normal-operating-range batches.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Afucosylation	H	two_sided	2–13	7.98 ± 0.58	2.89
Galactosylation (total %Gal)	H	two_sided	10–40	26 ± 1.6	2.86
High Mannose	VH	two_sided	3–10	5.99 ± 0.51	1.94
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.1
Acidic charge variants (deamidation)	VL	upper	18–40	22.3 ± 1.4	4.26

The tightest capability at this step is 1.94 for High Mannose, whose simulated mean of 5.99 % sits 2.99 percentage points above its lower limit, and the lower limit is the binding side because the attribute is centred nearer to it than to its upper limit. That is the attribute of very high criticality at this step, which is why culture pH and culture temperature receive the closest control in §10. Galactosylation is next, at 2.86, with a mean of 26.0 % and 14.0 percentage points to its upper limit. The widest is 16.1 for Aggregates (HMW), which sits far below its upper limit under all operating conditions. Even the tightest of these capabilities clears its binding limit by the margin reported above.

The result is a robustness statement, and it is bounded in three ways. It holds for parameters drawn across their normal operating ranges and not across the wider characterization ranges. It is computed from the fitted response-surface models, so it inherits their uncertainty, and the model for acidic charge variants is the least well predicted of the five. It is computed in the qualified scale-down model, so it describes the capability the commercial process should show and not a capability measured at commercial scale. Confirmation at scale comes from the 5 process performance qualification batches.

The two process impurities attributed to this step are reported separately (Table 8.2), because their drug-substance capability is not this step's to claim.

Table 8.2: Drug-substance capability for the two process impurities generated at the production bioreactor. The clearance credit belongs to the purification steps.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.8

Host cell protein reaches a drug-substance capability of 6.14, and residual DNA is higher still. Neither figure is a property of the bioreactor, since the culture sets the input level and the purification train removes the material, with the principal credit at Protein A chromatography (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008). The bioreactor's contribution is to keep the input level consistent, which the classification of §9 and the monitoring of §10 are designed to do.

The tightest capability anywhere in the drug-substance process is 1.51, and it is not at this step. It belongs to the cumulative viral-clearance attributes consolidated in PCMR-001.

9 Parameter classification

Classification follows SOP-4001 and rests on two things: whether the parameter was shown to affect a critical quality attribute, and how much risk there is of the parameter leaving the design space in routine manufacture. A parameter linked to a critical attribute is critical, and it is called well controlled when the equipment and the control system hold it comfortably inside the region. The classifications are given in the parameter table (Table 4.1) and the reasons are as follows.

- Culture pH is a well-controlled critical process parameter, because it affects all five responses, carries the largest interactions in the study, and is the parameter on which the shape of the region most depends. It is classified as well controlled because the vessel holds pH continuously by carbon dioxide and base addition, and because the normal operating range covers 40.0% of the range studied.
- Culture temperature is a well-controlled critical process parameter, since it affects galactosylation, high mannose and acidic variants and participates in the excluded corner of §6. Temperature is the most reliably controlled variable in a jacketed bioreactor, and its operating range covers 50.0% of the characterized range.
- Dissolved carbon dioxide is a well-controlled critical process parameter, because it carries the largest single screening effect in the study, on acidic charge variants, and the second largest main effect on galactosylation. It is the least directly controlled of the five, since it is managed through sparge and overlay gas composition and agitation rather than by a dedicated actuator, which is why its characterized range was set widest and why it is monitored continuously in manufacture.
- Osmolality is a well-controlled critical process parameter, since it was significant for galactosylation and for acidic variants in screening, with the smallest main effects of the active factors on both, and it was not carried into the response-surface design. Its range is supported by the screening study and by the platform history, and it is held at scale by the feed schedule.
- Culture duration is a well-controlled critical process parameter, because it carries the largest effect on galactosylation and a large effect on aggregate, and it participates in the excluded corner of §6. Duration is set by a scheduled harvest decision, so the risk of an unplanned excursion is low, and the normal operating range covers 50.0% of the characterized range.
- Dissolved oxygen is a key process parameter, assessed univariately across 30 to 70 % of saturation and controlled to assure oxygen supply to the culture and consistent process performance.

- Initial viable cell concentration is a key process parameter, since it sets the starting point of the growth curve and with it the timing of the production phase, and it is controlled to keep batches comparable.
- Nutrient feed-1 volume is a key process parameter, because it determines nutrient availability through the production phase, and a shortfall in it was the subject of DEV-003-02 (§13).
- Basal medium concentration is a general process parameter, since it affects neither a critical quality attribute nor process consistency across the range studied, and it is controlled by the medium specification under SOP-2004.

The step therefore has 5 well-controlled critical process parameters, 3 key process parameters and 1 general process parameter, and no parameter classified as a critical process parameter. The absence of that class at this step reflects the control capability of the vessel and the size of the region, not an absence of quality-linked parameters. Across the whole drug-substance process there is 1 such parameter, and it is the inactivation pH of Step 6 (PCR-006).

10 Contribution to the control strategy

This step contributes the primary control for the glycan, charge-variant and aggregate attributes of the drug substance, and it contributes nothing to impurity or virus clearance. Its elements of the control strategy are the following.

The 5 well-controlled critical process parameters characterized here are held within the normal operating ranges of Table 4.1, which lie inside the design space of §6 with the margins reported there. Culture pH receives the closest control of the five, because it acts on every response and carries the interactions. Dissolved carbon dioxide is monitored continuously and trended, because it is the parameter whose control capability is least direct.

The 3 key process parameters are held within their ranges and monitored to assure process consistency, and they are not linked to a critical quality attribute. Nutrient feed delivery is reconciled by mass at the end of every batch, which is a control introduced after DEV-003-02 and described in §13. Probe performance is verified against an offline reference during each batch, which addresses DEV-003-01.

Release testing covers the five formed attributes on every batch of drug substance, by the validated methods of this step (Table 3.1). Host cell protein and residual DNA are also released attributes, but the assurance for them is provided by the purification steps and not here. The lifecycle controls for this step are the calibration and change control of the bioreactor instrumentation, the material specifications for medium and feed under SOP-2004, and continued process verification against the capability of §8.

The step does not remove host cell protein, DNA, leached Protein A or aggregate, and it makes no contribution to the viral safety of the product. The cumulative position on all of those is consolidated in PCMR-001, which draws on PCR-005 through PCR-010.

11 Discussion

The production bioreactor is now characterized to a level that supports the operating ranges proposed for commercial manufacture. The 4 parameters carried into the response-surface study account for the behaviour of all five formed attributes, the fitted models are adequate and free of lack of fit, and the mechanistic account in §5.4 agrees with the platform expectation set out before the study was run. Where the study adds most is in the interactions. Three of the four response-surface parameters interact with culture pH, and no set of univariate ranges reproduces the region those interactions define.

The single adverse finding is the excluded corner. Galactosylation exceeds its upper limit at the vertex where culture pH, dissolved carbon dioxide and culture duration are all low and temperature is high, by 0.27 percentage points on the model and by more in the runs where it was measured. Two considerations bound the finding, since reaching the vertex requires four simultaneous excursions to the edge of the knowledge space, and returning any one parameter by a small amount brings the response back inside the limit. Within the normal operating ranges the worst predicted galactosylation is 31.8 %, which leaves 8.2 percentage points of margin. The finding is retained as an edge of failure, and it is the reason the design space of §6 is stated as the characterized region less one corner instead of as the whole characterized region.

Confidence in the scale-down model is good for the questions this study asks and limited for others. The model reproduces the culture environment that forms the attributes, and the centre points of both designs repeat within the precision of the analytical methods. It cannot represent the mixing and gas-transfer heterogeneity of a 15,000 L vessel at full working volume, and dissolved carbon dioxide is the parameter for which that gap is most likely to matter. That is one of the reasons pCO_2 was characterized over the widest relative range and is monitored continuously in manufacture.

2 deviations occurred during execution and both were retained. The pCO_2 probe drift of DEV-003-01 displaced one screening run by less than the reproducibility of the responses it could have affected. The feed under-delivery of DEV-003-02 left the affected centre-point batch inside the normal operating range of the parameter concerned. Neither changed a fitted effect, a classification or the boundary of the region, and the analysis in §13 gives the magnitudes.

The principal limitations of the study are three. The design space is defined in a scale-down model and its edges are not confirmed at commercial scale, which is a general property of design spaces and is addressed by the qualification campaign and by continued process verification. The capability figures are model-based projections over the operating ranges and not measurements of commercial batches. Osmolality is supported by a screening study and by platform history and not by a response-surface model, which is a deliberate choice justified by the size of its main effects. It leaves the interaction between osmolality and culture temperature on galactosylation quantified only at the resolution of the screening design, and it constrains any future proposal to widen the osmolality range.

No parameter at this step required the most stringent control classification. Every quality-linked parameter is held by a dedicated controller or by a scheduled decision, and the operating ranges sit well inside the region, so the risk of leaving the design space in routine manufacture is low for all five.

12 Conclusions

The production bioreactor is well understood over the ranges characterized, and the study meets the objectives set in §1.2. A response-surface model in 4 parameters defines an operating region over which the glycan, charge-variant and aggregate attributes formed at this step meet their acceptance criteria, with one corner of the characterized region excluded on galactosylation grounds. The proven acceptable ranges of §7 span the full characterization range of every parameter against every governed attribute, so the binding constraint on operation is the multivariate corner and not any univariate range.

9 parameters were classified, 5 of them as well-controlled critical process parameters, and none required classification as a critical process parameter. At commercial scale every attribute formed at this step meets its criterion, with the tightest capability of 1.94 belonging to High Mannose and a margin of 2.99 percentage points to its binding limit. The 2 recorded deviations were investigated and retained without impact on the fitted models, the classifications or the region.

The step makes no impurity-clearance or virus-clearance claim, and the assurance for host cell protein, DNA and viral safety is provided by the purification steps and consolidated in PCMR-001. Confirmation of the region at commercial scale is the responsibility of the process performance qualification campaign.

13 Deviations from the plan

2 deviations from PCP-003 were recorded during execution. Each was investigated to root cause, impact-assessed against the study objectives and dispositioned before the analysis was locked. Both were retained, which means that the affected data were kept in the analysed data set and that the justification for keeping them is given below. The register is given below (Table 13.1).

Table 13.1: Deviations recorded during execution of PCP-003.

Deviation	Summary	Detected during	Disposition
DEV-003-01	Dissolved-CO2 probe drift during screening run 4	in-process monitoring vs offline reference	retained
DEV-003-02	Nutrient feed-1 under-delivery on a centre-point batch	post-execution mass reconciliation	retained

Both deviations concern the delivery of a set condition to the culture rather than the measurement of a response, and both were detected by a control that was already in place. Neither was detected by the product-quality result, which is the weaker way to find a deviation of this kind.

13.1 DEV-003-01 — dissolved carbon dioxide probe drift

The in-process pCO_2 probe on screening run 4 read 6.5 mmHg away from the offline reference measurement. The run was controlled to its target on the probe reading, so the culture experienced a pCO_2 displaced from the nominal level of the design point by that amount. The nominal pCO_2 for that run was 160 mmHg.

The investigation identified drift of the probe between scheduled calibrations as the root cause. The instrument concerned is EQ-BRX-205, the production scale-down bioreactor pCO2 probe, which was within its calibration interval at execution and next due on 2026-06-14. No other run of either design showed a discrepancy against the offline reference.

The impact assessment used the fitted screening model, in which a displacement of 6.5 mmHg is 0.11 in coded units, about one ninth of the half-range of the factor. Evaluating the screening model at the settings of run 4 with and without that displacement changes the most sensitive response, Acidic variants, by 0.40 percentage points. The centre-point standard deviation of that response in the same design is 1.09 percentage points, so the displacement moves the run by less than the study can resolve. The deviation was retained on that basis. The corresponding control for manufacture is the verification of probe readings against an offline reference during each batch, described in §10.

13.2 DEV-003-02 — nutrient feed-1 under-delivery

Post-execution mass reconciliation showed that the screening centre point batch received 4.2 % less nutrient feed-1 than the protocol specified. The material was nutrient feed-1 medium lot LOT-FED-3120 from Fern Scientific, and the shortfall was in the delivered volume and not in the composition of the feed.

The investigation identified a calibration offset on the feed pump as the root cause, confirmed by gravimetric check of the pump after the run. The offset was in one direction and was present for the affected batch only, since the pump was recalibrated before the remaining runs.

The impact assessment turned on where the delivered volume fell. The nutrient feed-1 set-point is 12.0 % of working volume, and a shortfall of 4.2 % delivered 11.50 % of working volume, which is inside the normal operating range of 10 to 14 %. The affected batch ran at a feed volume the process is qualified to run at, and the parameter is classified as a key process parameter with no demonstrated link to a critical quality attribute (§9).

The affected batch is a centre point, so its influence falls on the pure-error term rather than on any factor effect. 3 verification runs at the centre point were executed after the pump was recalibrated, and the glycan responses were measured by AMV-3010. The relative half-width of the 95 % confidence interval of that verification set is 6.21 %, which is the precision with which the centre-point mean is known. The deviation was retained on that basis, and the batch was kept in the analysed data set. Feed mass reconciliation at the end of every batch has been added to the control strategy for manufacture (§10).

14 Appendices A–D

The appendices carry the full design matrices, the complete effect and coefficient tables, and the analytical method summary. They are the primary record behind the tables shown in §5.

14.1 Appendix A — Screening design matrix and responses

The coded settings of the screening design are given first (Table 14.1), followed by the responses obtained (Table 14.2). Coded levels are minus one and plus one at the edges of each characterization range and zero at the set-point.

Table 14.1: Screening design, coded factor settings.

Run	Type	A	B	C	D	E
1	factorial	-1	-1	-1	-1	1
2	factorial	-1	-1	-1	1	-1
3	factorial	-1	-1	1	-1	-1
4	factorial	-1	-1	1	1	1
5	factorial	-1	1	-1	-1	-1
6	factorial	-1	1	-1	1	1
7	factorial	-1	1	1	-1	1
8	factorial	-1	1	1	1	-1
9	factorial	1	-1	-1	-1	-1
10	factorial	1	-1	-1	1	1
11	factorial	1	-1	1	-1	1
12	factorial	1	-1	1	1	-1
13	factorial	1	1	-1	-1	1
14	factorial	1	1	-1	1	-1
15	factorial	1	1	1	-1	-1
16	factorial	1	1	1	1	1
17	center	0	0	0	0	0
18	center	0	0	0	0	0
19	center	0	0	0	0	0

Table 14.2: Screening design, measured responses.

Run	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
1	5.18	27.9	6.61	26.4	1.3
2	8.93	36.1	6.44	26.5	0.783
3	11.8	34.6	5.63	22.3	0.785
4	4.83	14.7	5.46	17.9	1.48
5	9.72	41.4	4.61	27	0.859
6	4.68	27.3	4.68	27.6	1.26
7	6.22	14.5	4.65	23.1	1.48
8	9.83	31.2	4.45	19.3	0.948
9	9.61	28	7.77	24.1	1.51
10	7.51	17	7.89	20.4	2.26
11	7.13	19.5	7.34	12.6	2.53
12	5.29	17.6	7.01	11.3	1.61
13	6.74	28.3	6.43	29.6	2.18
14	5.4	26.7	5.94	24.4	1.62
15	3.82	25.2	6.1	18.4	1.8
16	3.08	29.1	6.15	15.8	2.05
17	7.8	26.4	5.88	21.9	1.64
18	7.51	27.4	5.84	20	1.63
19	8.48	27	7.11	21.8	1.5

14.2 Appendix B — Response-surface design matrix and responses

The coded settings of the face-centred central composite design are given first (Table 14.3), followed by the responses obtained (Table 14.4).

Table 14.3: Response-surface design, coded factor settings.

Run	Type	A	B	E	C
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1

Run	Type	A	B	E	C
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 14.4: Response-surface design, measured responses.

Run	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
1	9.43	38.8	5.44	29.8	0.827
2	11.8	29	5.55	17.6	0.791
3	4.5	25.9	5.65	25.1	1.33
4	5.38	16.6	5.38	20	1.37
5	10.4	40.5	4.23	25.6	0.803
6	10.1	30.7	4.17	19.8	0.707
7	5.12	27.1	4.63	30.3	1.24
8	6.13	17.4	4.64	21.8	1.41
9	8.92	22	6.79	22.9	1.5
10	5.89	18.7	8.81	13.6	1.53
11	8.05	18.7	6.11	19.3	1.99
12	5.17	18.7	7.34	12.3	2.15
13	5.41	28.7	6.97	28.5	1.45
14	2.84	29.1	7.36	18.5	1.7
15	6.2	25.4	5.95	26.5	2.08
16	4.19	27.2	6.8	17.2	2.01
17	7.82	26.1	4.59	24.2	1.15
18	6.07	23.7	7.72	19.3	1.82
19	8.77	24.3	6.75	19.9	1.51
20	7.55	27.9	5.51	26.8	1.32
21	10.3	30.8	6.11	20.8	1.3
22	6.89	22.5	5.54	22.4	2.06

Run	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
23	8.48	27.1	6.39	25.7	1.5
24	8.39	23.5	5.94	19.2	1.45
25	7.88	23	5.75	20.5	1.52
26	7.95	27.3	6.32	22	1.6
27	7.35	25	5.6	22.4	1.55
28	7.27	28.2	4.96	21.1	1.37

14.3 Appendix C — Full effect and coefficient tables

The screening model summaries for all five responses are given first (Table 14.5), and the complete effect estimates for each response follow. The response-surface coefficient tables for the three responses not shown in §5.3 complete the appendix.

Table 14.5: Screening model summaries for all responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Afucosylation	19	0.965	0.79	-47.4	5.51	0.0926	1.06
Galactosylation	19	0.998	0.988	-1.33	103	0.00138	0.773
High mannose	19	0.942	0.65	-7.99	3.23	0.182	0.619
Acidic variants	19	0.994	0.963	-0.442	32.2	0.00766	0.957
Aggregates (HMW)	19	0.995	0.972	-2.44	42.2	0.00515	0.0829

Table 14.6: Screening effect estimates for afucosylation.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AE	2.46	1.23	0.266	4.62	0.0191	*
E	-2.37	-1.19	0.266	-4.46	0.0209	*
AC	-1.76	-0.88	0.266	-3.31	0.0455	*
A	-1.57	-0.785	0.266	-2.95	0.06	
B	-1.34	-0.671	0.266	-2.53	0.0858	
D	-1.33	-0.664	0.266	-2.5	0.0878	
AB	-1.28	-0.64	0.266	-2.41	0.0952	
C	-0.73	-0.365	0.266	-1.37	0.264	
BD	0.452	0.226	0.266	0.85	0.458	
BE	0.362	0.181	0.266	0.681	0.545	
AD	-0.175	-0.0876	0.266	-0.329	0.764	
BC	-0.171	-0.0856	0.266	-0.322	0.769	
CD	-0.147	-0.0736	0.266	-0.277	0.8	
DE	0.0358	0.0179	0.266	0.0674	0.951	
CE	0.0118	0.00591	0.266	0.0222	0.984	

Table 14.7: Screening effect estimates for galactosylation.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
E	-7.82	-3.91	0.193	-20.2	0.000264	***
AE	6.9	3.45	0.193	17.9	0.000382	***
C	-5.79	-2.9	0.193	-15	0.000644	***
A	-4.56	-2.28	0.193	-11.8	0.00131	**
BD	3.7	1.85	0.193	9.57	0.00242	**
AC	3.6	1.8	0.193	9.32	0.00261	**
B	3.54	1.77	0.193	9.17	0.00274	**
AB	3.28	1.64	0.193	8.49	0.00343	**
D	-2.45	-1.23	0.193	-6.35	0.0079	**
CD	2.18	1.09	0.193	5.64	0.011	*
DE	1.93	0.963	0.193	4.98	0.0155	*
BE	1.49	0.747	0.193	3.87	0.0306	*
AD	-0.195	-0.0974	0.193	-0.504	0.649	
BC	-0.131	-0.0657	0.193	-0.34	0.756	
CE	0.119	0.0596	0.193	0.309	0.778	

Table 14.8: Screening effect estimates for high mannose.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.51	0.756	0.155	4.89	0.0164	*
B	-1.39	-0.696	0.155	-4.5	0.0205	*
C	-0.446	-0.223	0.155	-1.44	0.245	
BC	0.369	0.185	0.155	1.19	0.318	
E	0.158	0.0791	0.155	0.511	0.645	
D	-0.139	-0.0693	0.155	-0.448	0.685	
AC	0.0897	0.0449	0.155	0.29	0.791	
AE	0.0894	0.0447	0.155	0.289	0.792	
DE	-0.0718	-0.0359	0.155	-0.232	0.831	
CE	-0.0552	-0.0276	0.155	-0.179	0.87	
BE	0.046	0.023	0.155	0.149	0.891	
AB	0.0432	0.0216	0.155	0.14	0.898	
AD	-0.0239	-0.012	0.155	-0.0774	0.943	
CD	-0.0232	-0.0116	0.155	-0.075	0.945	
BD	-0.000576	-0.000288	0.155	-0.00186	0.999	

Table 14.9: Screening effect estimates for acidic variants.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-8.15	-4.08	0.239	-17	0.000441	***
A	-4.17	-2.08	0.239	-8.7	0.00319	**
B	2.97	1.48	0.239	6.19	0.00848	**

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	-2.55	-1.28	0.239	-5.34	0.0129	*
AB	1.98	0.992	0.239	4.15	0.0255	*
AC	-1.93	-0.964	0.239	-4.03	0.0275	*
BE	1.73	0.865	0.239	3.61	0.0364	*
AD	-0.669	-0.335	0.239	-1.4	0.256	
CE	-0.494	-0.247	0.239	-1.03	0.378	
CD	-0.492	-0.246	0.239	-1.03	0.379	
BD	-0.215	-0.108	0.239	-0.449	0.684	
BC	0.151	0.0757	0.239	0.316	0.773	
DE	0.0324	0.0162	0.239	0.0676	0.95	
E	0.0235	0.0117	0.239	0.049	0.964	
AE	0.0226	0.0113	0.239	0.0472	0.965	

Table 14.10: Screening effect estimates for aggregates (hmw).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.834	0.417	0.0207	20.1	0.000268	***
E	0.578	0.289	0.0207	13.9	0.0008	***
BE	-0.141	-0.0705	0.0207	-3.4	0.0424	*
C	0.116	0.058	0.0207	2.8	0.0678	
CD	-0.0742	-0.0371	0.0207	-1.79	0.171	
AD	-0.0666	-0.0333	0.0207	-1.61	0.206	
AB	-0.0593	-0.0297	0.0207	-1.43	0.248	
DE	-0.0554	-0.0277	0.0207	-1.34	0.274	
D	-0.0548	-0.0274	0.0207	-1.32	0.278	
BD	-0.0547	-0.0274	0.0207	-1.32	0.278	
AE	0.0416	0.0208	0.0207	1	0.39	
BC	-0.0235	-0.0118	0.0207	-0.567	0.61	
CE	0.0222	0.0111	0.0207	0.537	0.629	
B	-0.00671	-0.00336	0.0207	-0.162	0.882	
AC	-0.00663	-0.00332	0.0207	-0.16	0.883	

Table 14.11: Response-surface coefficients for afucosylation.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	8.04	0.202	39.8	5.74e-15	***
A	-0.997	0.138	-7.24	6.52e-06	***
B	-0.551	0.138	-4	0.00151	**
E	-1.3	0.138	-9.45	3.45e-07	***
C	-0.367	0.138	-2.67	0.0194	*
AB	-0.63	0.146	-4.32	0.000837	***
AE	1.32	0.146	9.06	5.6e-07	***
AC	-0.907	0.146	-6.21	3.16e-05	***

Term	Coef.	Std. err.	t	p-value	Sig.
BE	0.362	0.146	2.48	0.0277	*
BC	-0.0703	0.146	-0.481	0.638	
EC	0.0322	0.146	0.221	0.829	
A^2	-1.37	0.364	-3.77	0.00233	**
B^2	-0.157	0.364	-0.431	0.673	
E^2	0.257	0.364	0.705	0.493	
C^2	0.117	0.364	0.323	0.752	

Table 14.12: Response-surface coefficients for acidic variants.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	22	0.542	40.6	4.43e-15	***
A	-2.01	0.369	-5.44	0.000113	***
B	1.9	0.369	5.14	0.000191	***
E	-0.124	0.369	-0.335	0.743	
C	-4.09	0.369	-11.1	5.4e-08	***
AB	1.11	0.392	2.84	0.0139	*
AE	-0.793	0.392	-2.03	0.0638	
AC	-0.248	0.392	-0.633	0.538	
BE	0.655	0.392	1.67	0.118	
BC	-0.0111	0.392	-0.0283	0.978	
EC	0.465	0.392	1.19	0.256	
A^2	-0.586	0.975	-0.601	0.558	
B^2	1.03	0.975	1.06	0.311	
E^2	-0.718	0.975	-0.736	0.475	
C^2	0.13	0.975	0.133	0.896	

Table 14.13: Response-surface coefficients for aggregates (hmw).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	1.52	0.0341	44.8	1.25e-15	***
A	0.367	0.0232	15.8	7.26e-10	***
B	-0.0157	0.0232	-0.675	0.512	
E	0.281	0.0232	12.1	1.91e-08	***
C	0.0225	0.0232	0.971	0.349	
AB	0.0135	0.0246	0.547	0.593	
AE	-0.0106	0.0246	-0.43	0.674	
AC	0.0179	0.0246	0.727	0.48	
BE	-0.00713	0.0246	-0.29	0.777	
BC	0.00414	0.0246	0.168	0.869	
EC	0.0105	0.0246	0.426	0.677	
A^2	-0.0533	0.0613	-0.87	0.4	
B^2	-0.119	0.0613	-1.93	0.0752	

Term	Coef.	Std. err.	t	p-value	Sig.
E^2	0.143	0.0613	2.33	0.0366	*
C^2	-0.0626	0.0613	-1.02	0.326	

14.4 Appendix D — Analytical methods summary

The validated methods used for the responses and the released attributes of this step are listed below (Table 14.14). Method performance is summarized here and reported in full in the validation reports.

Table 14.14: Validated analytical methods supporting this report.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

The released N-glycan map (AMV-3010) reports afucosylation, galactosylation and high mannose from one separation, with a relative precision of 2.5 %, an accuracy of 99.0 % and a limit of quantitation of 0.5 % of glycans. The host cell protein ELISA (AMV-3012) has a relative precision of 9.5 %, an accuracy of 95.0 % and a limit of quantitation of 1.0 ng/mg. Size-variant analysis (AMV-3011), charge-variant analysis (AMV-3013) and residual DNA (AMV-3014) are qualified for their reportable ranges in their own validation reports, which are the location of record for their performance characteristics.

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